Industrial Internship Report on

" E-Commerce Website for Automotive Parts"

Prepared by

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Executive Summary

This report provides details of the Industrial Internship provided by upskill Campus and The IoT Academy in collaboration with Industrial Partner UniConverge Technologies Pvt Ltd (UCT).

This internship was focused on a project/problem statement provided by UCT. We had to finish the project including the report in 6 weeks' time.

My project was an e-commerce website for automotive parts called PartWorks. The idea was to build a platform where users can browse through different automotive parts and vehicles, add the ones they need to their cart, and check out easily. I also added features like user login and account management, an admin panel so the store owner can manage products and orders, an affiliate section, and a deals page for offers. For the frontend, I used React with TypeScript, and Vite for the build setup. I broke the project down into separate pages for the catalog, cart, checkout, account, and admin sections so it would be easier to manage and scale. One feature I'm particularly happy with is the vehicle selector tool, which helps users quickly find parts that fit their specific vehicle.

This internship gave me a very good opportunity to get exposure to Industrial problems and design/implement solution for that. It was an overall great experience to have this internship.

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1 Preface

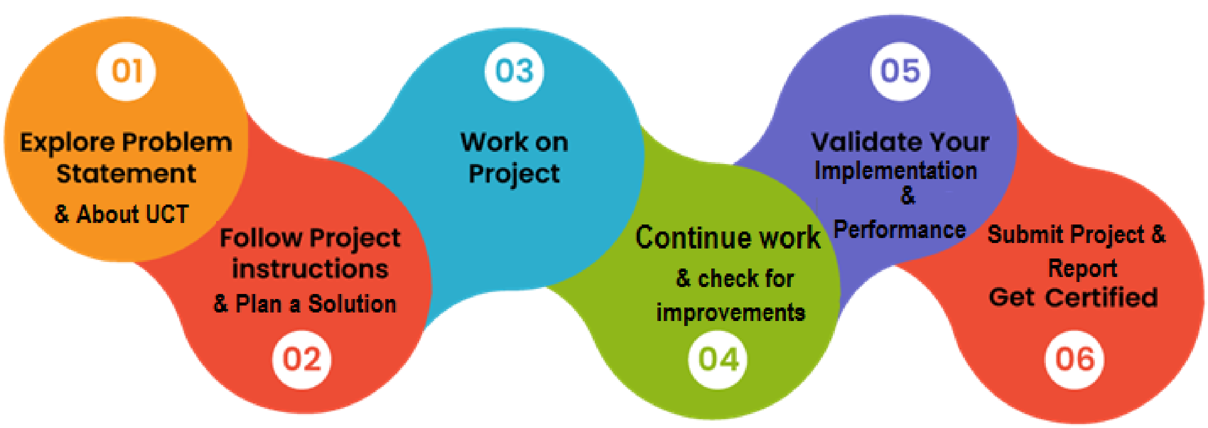
Over the course of this 6-week industrial internship, I designed and developed PartWorks, an e-commerce website for automotive parts, as my assigned project from UniConverge Technologies Pvt Ltd (UCT). The work covered the full cycle of planning, building, and documenting a real-world web application, from the initial problem statement through to a working product.

Internships like this one are extremely valuable in bridging the gap between academic learning and real industry expectations. Working on an actual problem statement, within a fixed timeline, gave me exposure to how projects are scoped, built, and delivered in a professional setting — something classroom coursework alone does not fully prepare you for.

My project, PartWorks, is an e-commerce website for automotive parts. It allows customers to browse a catalog of parts and vehicles, use a vehicle selector tool to find compatible parts, add items to a cart, and complete purchases through a checkout flow. It also includes an admin panel for managing products and orders, an affiliate section, and a deals/offers page.

I am grateful to upSkill Campus, The IoT Academy, and UniConverge Technologies Pvt Ltd (UCT) for the opportunity to work on this internship and gain hands-on, industry-oriented project experience.

The program was structured over 6 weeks, starting with an introduction to the problem statement and requirements, followed by design and planning of the application, iterative development of the core features, testing, and finally documentation of the work in this report.



Overall, this internship was a great learning experience. I improved my skills in React and TypeScript, learned how to structure a multi-page application, and got a better understanding of the full product lifecycle — from a problem statement to a working, documented solution.

I would like to thank the mentors and coordinators at upSkill Campus, The IoT Academy, and UniConverge Technologies Pvt Ltd (UCT) for their guidance and support throughout this internship.

To juniors and peers taking up this internship: start early, break your project down into small manageable pieces, and don’t hesitate to explore beyond the minimum requirements — it makes the final result, and the learning, much stronger.

2 Introduction

2.1 About UniConverge Technologies Pvt Ltd

A company established in 2013 and working in Digital Transformation domain and providing Industrial solutions with prime focus on sustainability and RoI.

For developing its products and solutions it is leveraging various **Cutting Edge Technologies e.g. Internet of Things (IoT), Cyber Security, Cloud computing (AWS, Azure), Machine Learning, Communication Technologies (4G/5G/LoRaWAN), Java Full Stack, Python, Front end** etc.

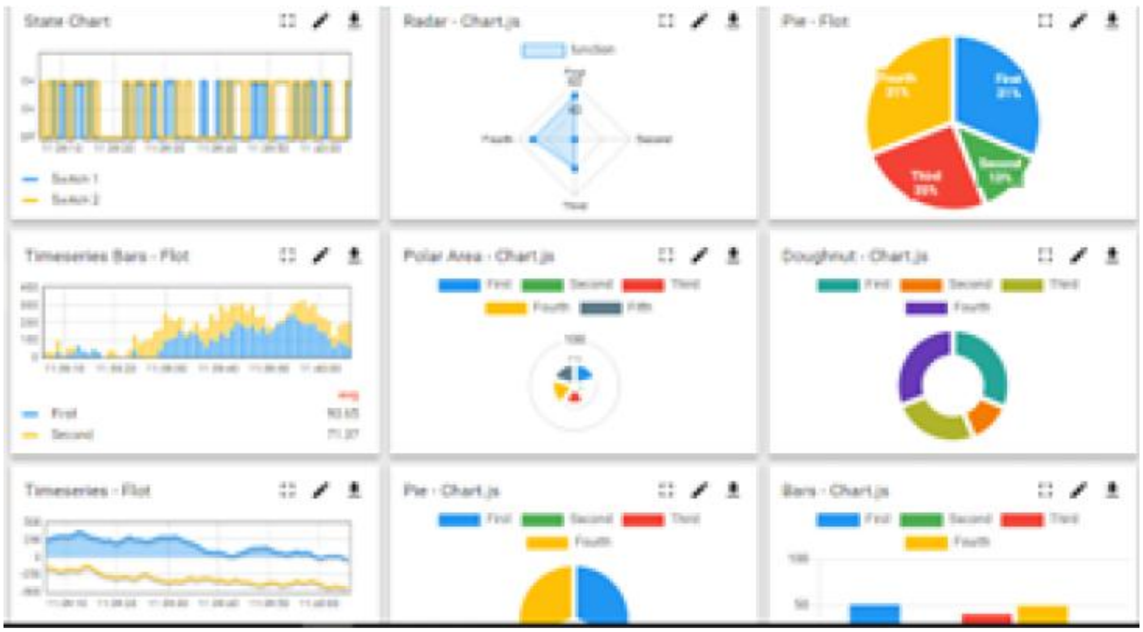
i. UCT IoT Platform ( )

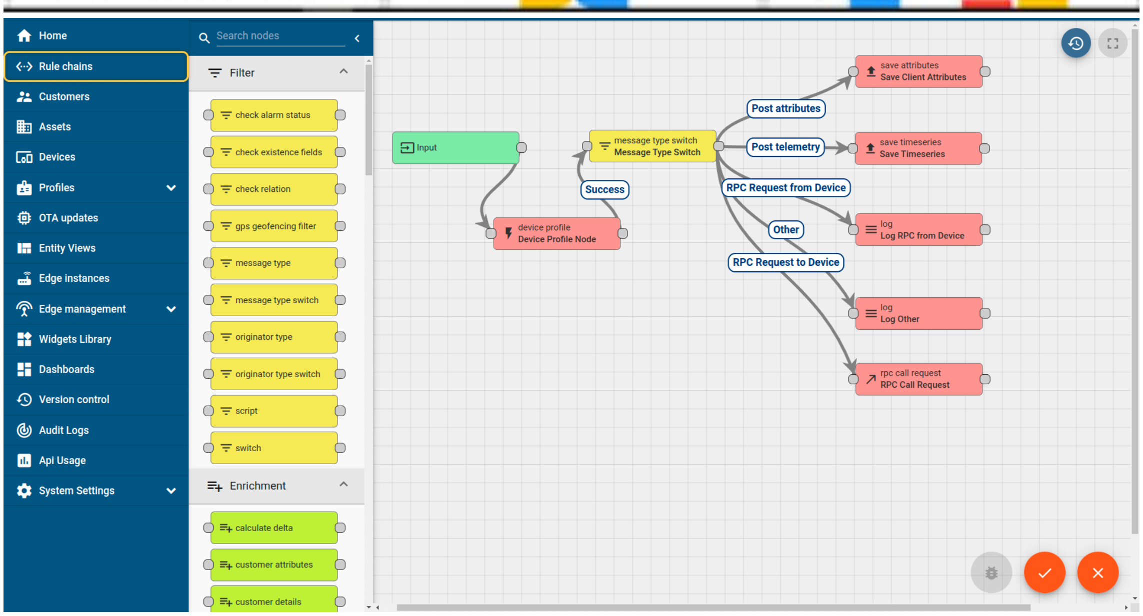
UCT Insight is an IOT platform designed for quick deployment of IOT applications on the same time providing valuable "insight" for your process/business. It has been built in Java for backend and ReactJS for Front end. It has support for MySQL and various NoSql Databases.

- It enables device connectivity via industry standard IoT protocols - MQTT, CoAP, HTTP, Modbus TCP, OPC UA
- It supports both cloud and on-premises deployments.

It has features to

- Build Your own dashboard
- Analytics and Reporting
- Alert and Notification
- Integration with third party application(Power BI, SAP, ERP)
- Rule Engine





ii. Smart Factory Platform (FACTORY WATCH)

Factory watch is a platform for smart factory needs.

It provides Users/ Factory

- with a scalable solution for their Production and asset monitoring
- OEE and predictive maintenance solution scaling up to digital twin for your assets.
- to unleased the true potential of the data that their machines are generating and helps to identify the KPIs and also improve them.
- A modular architecture that allows users to choose the service that they what to start and then can scale to more complex solutions as per their demands.

Its unique SaaS model helps users to save time, cost and money.

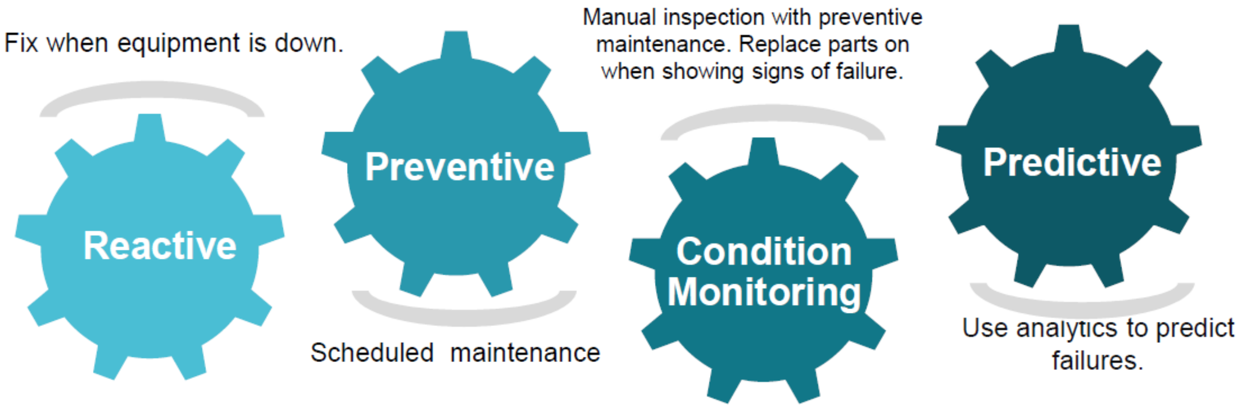
Machine	Operator	Work Order ID	Job ID	Job Performance	Job Progress		Output		Rejection	Time (mins)				Job Status	End Customer
					Start Time	End Time	Planned	Actual		Setup	Pred	Downtime	Idle		
CNC_S7_81	Operator 1	WO0405200001	4168	58%	10:30 AM		55	41	0	80	215	0	45	In Progress	i
CNC_S7_81	Operator 1	WO0405200001	4168	58%	10:30 AM		55	41	0	80	215	0	45	In Progress	i

iii. LoRaWAN based Solution

UCT is one of the early adopters of LoRAWAN teschnology and providing solution in Agritech, Smart cities, Industrial Monitoring, Smart Street Light, Smart Water/ Gas/ Electricity metering solutions etc.

iv. Predictive Maintenance

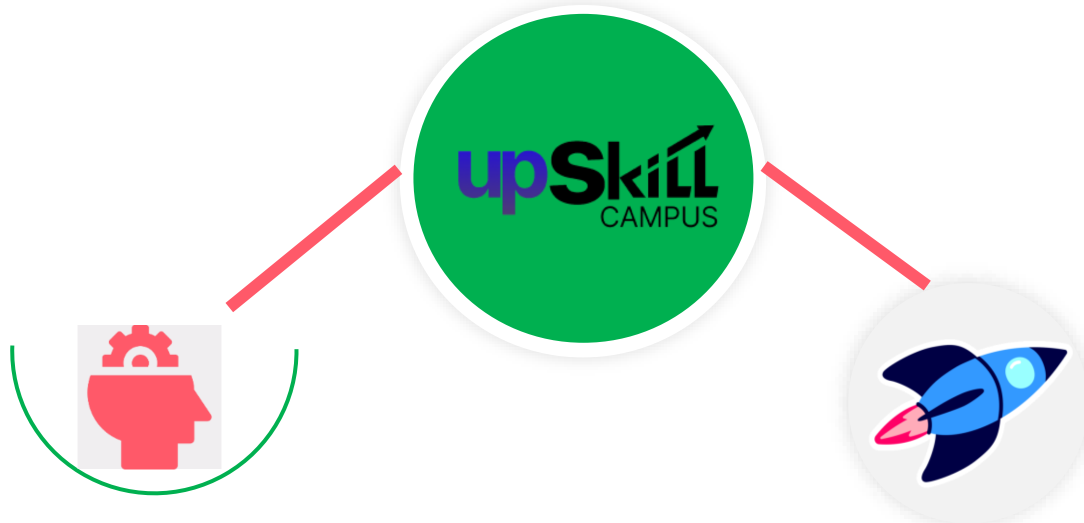
UCT is providing Industrial Machine health monitoring and Predictive maintenance solution leveraging Embedded system, Industrial IoT and Machine Learning Technologies by finding Remaining useful life time of various Machines used in production process.



2.2 About upskill Campus (USC)

upskill Campus along with The IoT Academy and in association with Uniconverge technologies has facilitated the smooth execution of the complete internship process.

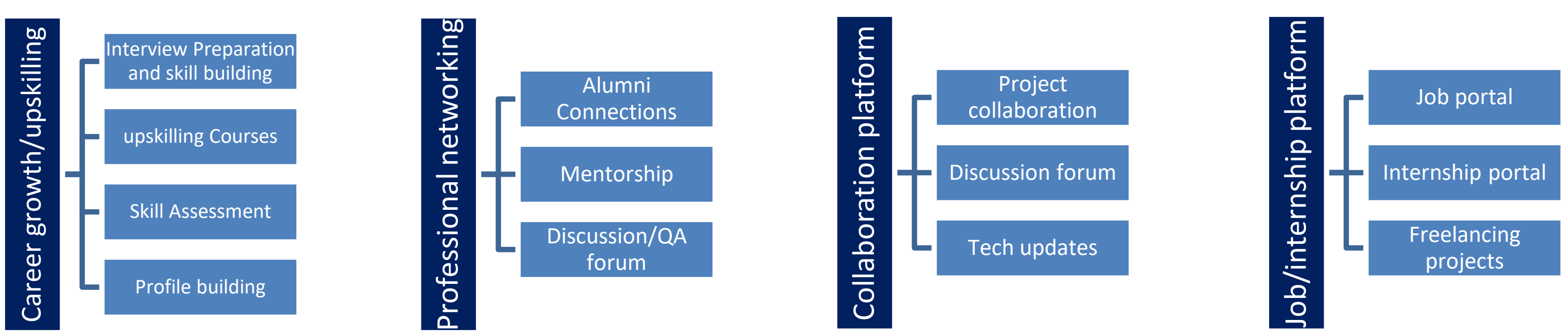
USC is a career development platform that delivers **personalized executive coaching** in a more affordable, scalable and measurable way.



Seeing need of upskilling in self paced manner along-with additional support services e.g. Internship, projects, interaction with Industry experts, Career growth Services

upSkill Campus aiming to upskill 1 million learners in next 5 year

https://www.upskillcampus.com/



2.3 The IoT Academy

The IoT academy is EdTech Division of UCT that is running long executive certification programs in collaboration with EICT Academy, IITK, IITR and IITG in multiple domains.

2.4 Objectives of this Internship program

The objective for this internship program was to

☛ get practical experience of working in the industry.

☛ to solve real world problems.

☛ to have improved job prospects.

☛ to have Improved understanding of our field and its applications.

☛ to have Personal growth like better communication and problem solving.

2.5 Reference

[1] React Documentation – https://react.dev

[2] TypeScript Documentation – https://www.typescriptlang.org/docs

[3] Vite Documentation – https://vitejs.dev

2.6 Glossary

Terms	Acronym
UCT	UniConverge Technologies Pvt Ltd
IoT	Internet of Things
UI/UX	User Interface / User Experience
SPA	Single Page Application
API	Application Programming Interface

3 Problem Statement

In the assigned problem statement

Customers looking to purchase automotive spare parts often struggle to find parts that are compatible with their specific vehicle, and many small or local automotive parts sellers lack an organized online presence. There was a need for a web-based platform that lets customers browse a categorized catalog of automotive parts, check vehicle compatibility, and complete purchases online — while also giving the business owner an admin interface to manage inventory and orders.

4 Existing and Proposed solution

Most small automotive parts retailers rely on manual, in-store sales or generic marketplace listings that don't offer vehicle-specific compatibility checks, making it hard for customers to be sure a part will fit their vehicle before purchasing. There is also often no unified system covering both customer purchases and admin-side inventory or order management.

PartWorks is a dedicated e-commerce platform for automotive parts that addresses this gap. It provides a searchable and filterable product catalog, a vehicle selector tool to match parts to specific vehicles, a shopping cart and checkout flow, user account management, and an admin panel for managing products and orders.

The main value addition is bringing the entire buying and selling process — browsing, vehicle-based compatibility search, purchasing, and order/inventory management — into a single, structured web application, rather than relying on manual processes or generic, non-specialized marketplaces.

4.1 Code submission (Github link)

https://github.com/Lakshmi410799/upskillcampus

4.2 Report submission (Github link)

https://github.com/Lakshmi410799/upskillcampus/blob/main/PartWorks_Lakshmi_USC_UCT.pdf

5 Proposed Design/ Model

PartWorks follows a component-based frontend architecture built with React and TypeScript, using Vite as the build tool. The application flow starts with the customer landing on the homepage, moving through the catalog and vehicle selector to find a part, adding it to the cart, and completing checkout. Application-wide data such as cart contents and user session information is managed using React Context, so it stays consistent across pages without having to pass it manually through every component.

5.1 High Level Diagram (if applicable)

At a high level, the system consists of a React/TypeScript frontend organized into pages for Home, Catalog, Product Detail, Cart, Checkout, Account, Login, Order Confirmation, Deals, Affiliate, and Admin. Product and vehicle data currently flows from local data files (mockProducts.ts, mockVehicles.ts) into these pages, with room to extend this to a live backend/API in the future.

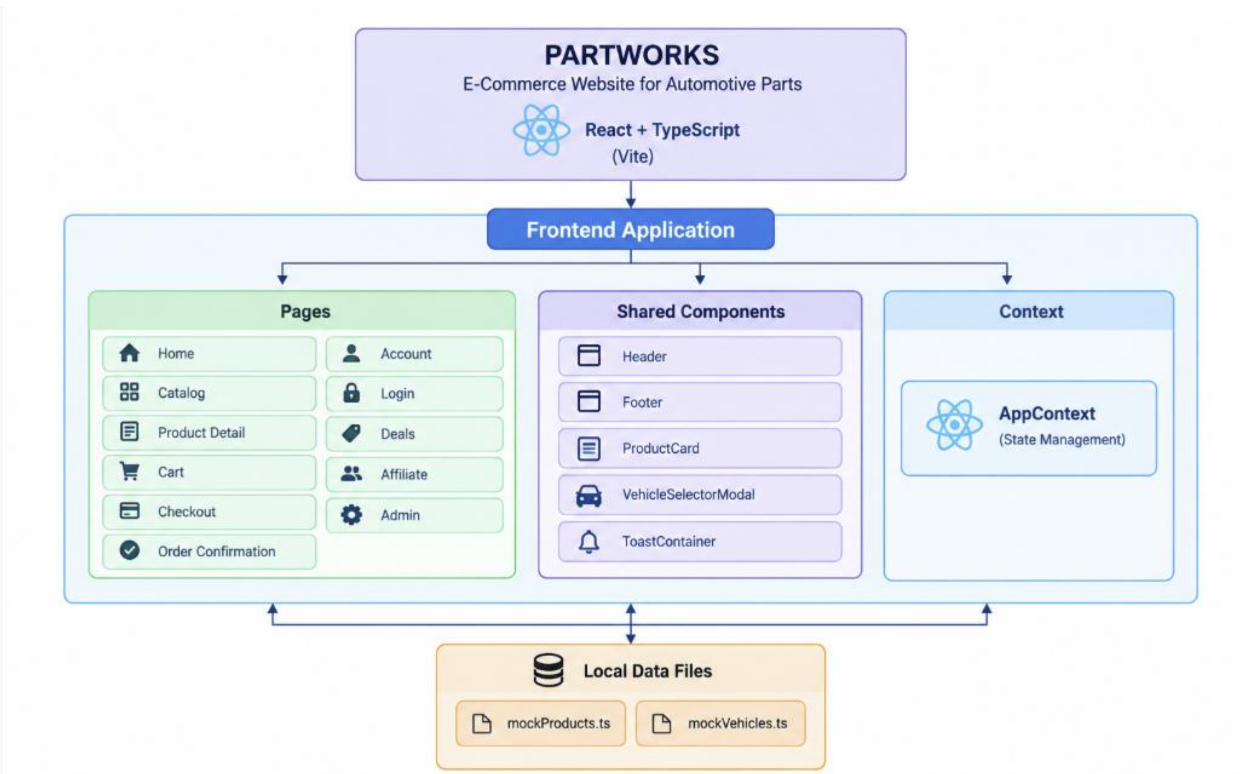
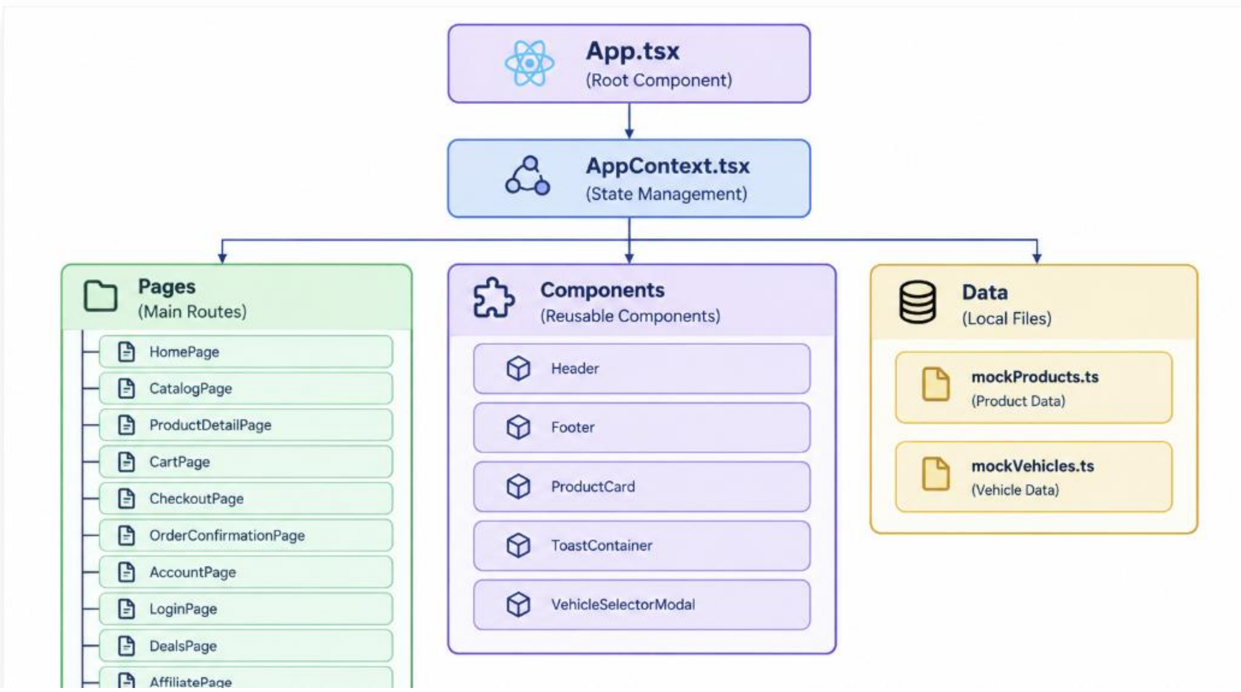


Figure 1: HIGH LEVEL DIAGRAM OF THE SYSTEM

5.2 Low Level Diagram (if applicable)



5.3 Interfaces (if applicable)

The system is structured into a Customer Interface (catalog browsing, vehicle selector, cart, checkout, login/account, order confirmation), an Admin Interface (product and order management), and a Data Interface (product and vehicle data feeding into the catalog and vehicle selector components). At the component level, reusable elements such as Header, Footer, ProductCard, VehicleSelectorModal, and ToastContainer feed into the page-level components, with mockProducts.ts and mockVehicles.ts (or an API, in a future iteration) as the underlying data source.

6 Performance Test

This section covers how PartWorks was tested to make sure it works reliably for real users, rather than just functioning as a one-off academic demo.

The main constraints for this project were functional correctness (core flows like cart and checkout must work reliably), usability (the interface needed to be intuitive across devices), and responsiveness (the layout should adapt properly to different screen sizes).

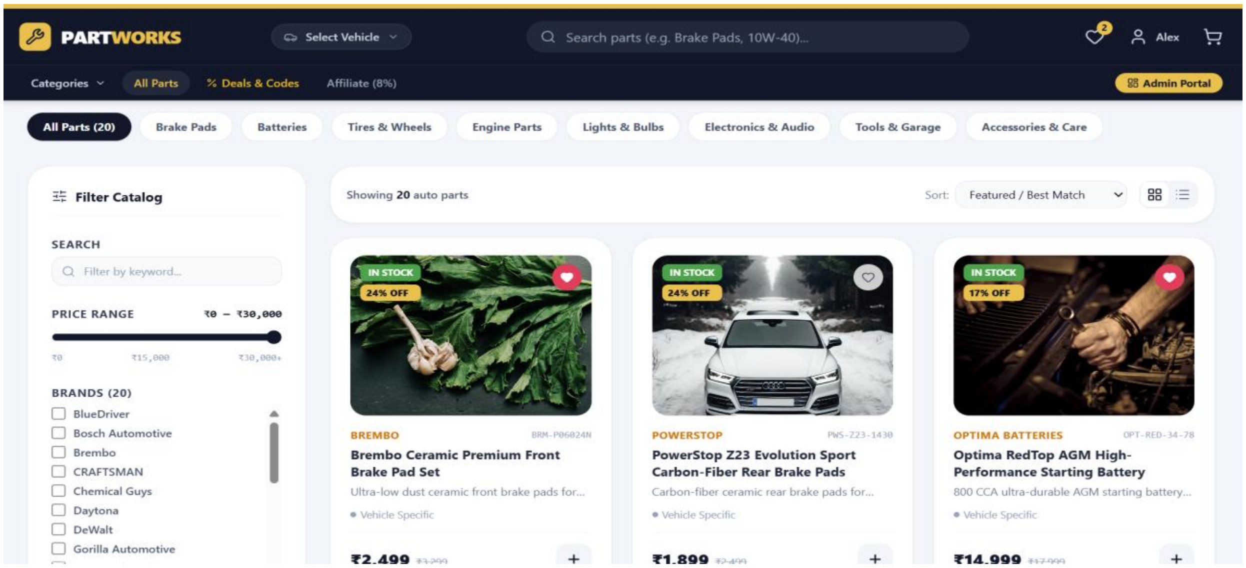
These constraints were addressed by structuring the application into clear, testable components, using React state/Context to keep cart and user data consistent, and building the UI with responsive layouts from the start rather than adapting it afterward.

Manual testing across the core user flows (browsing, vehicle selection, cart, checkout, login, and admin actions) confirmed that each worked as expected, and the layout adapted correctly across mobile and desktop screen sizes. Details of the specific test cases are listed below.

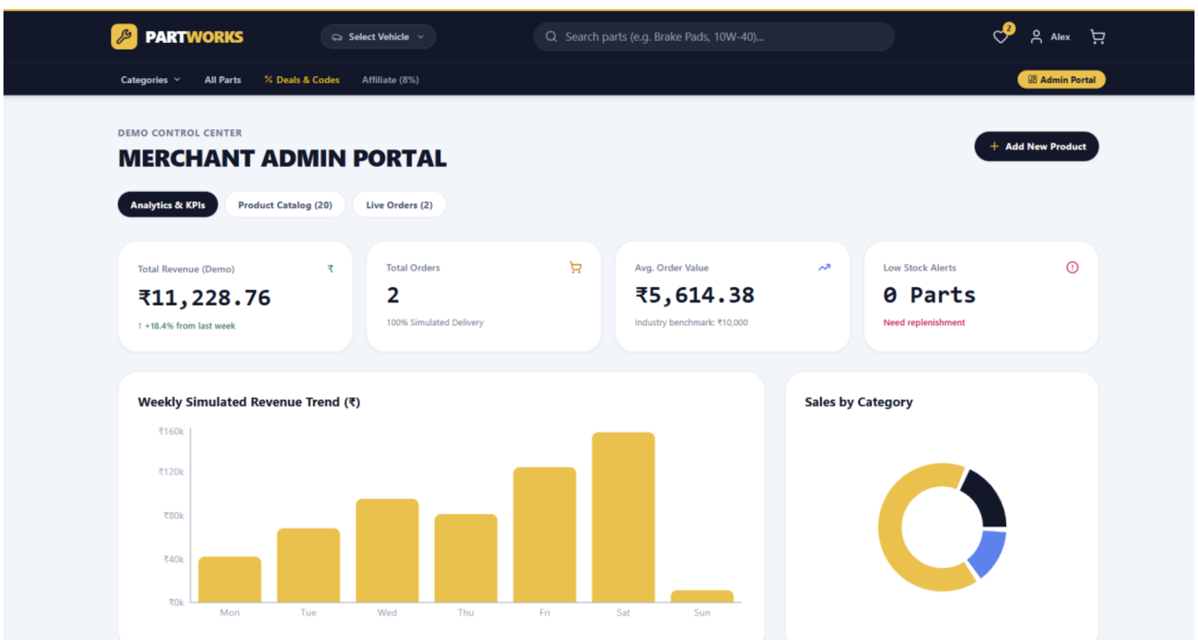
Since PartWorks is a web application rather than an embedded system, the relevant constraints were closer to usability, responsiveness, and functional correctness rather than memory or power consumption.

Formal automated/load testing was not carried out within the internship timeline; this is listed as a recommendation for future work, along with backend performance testing once a live API and database are integrated.

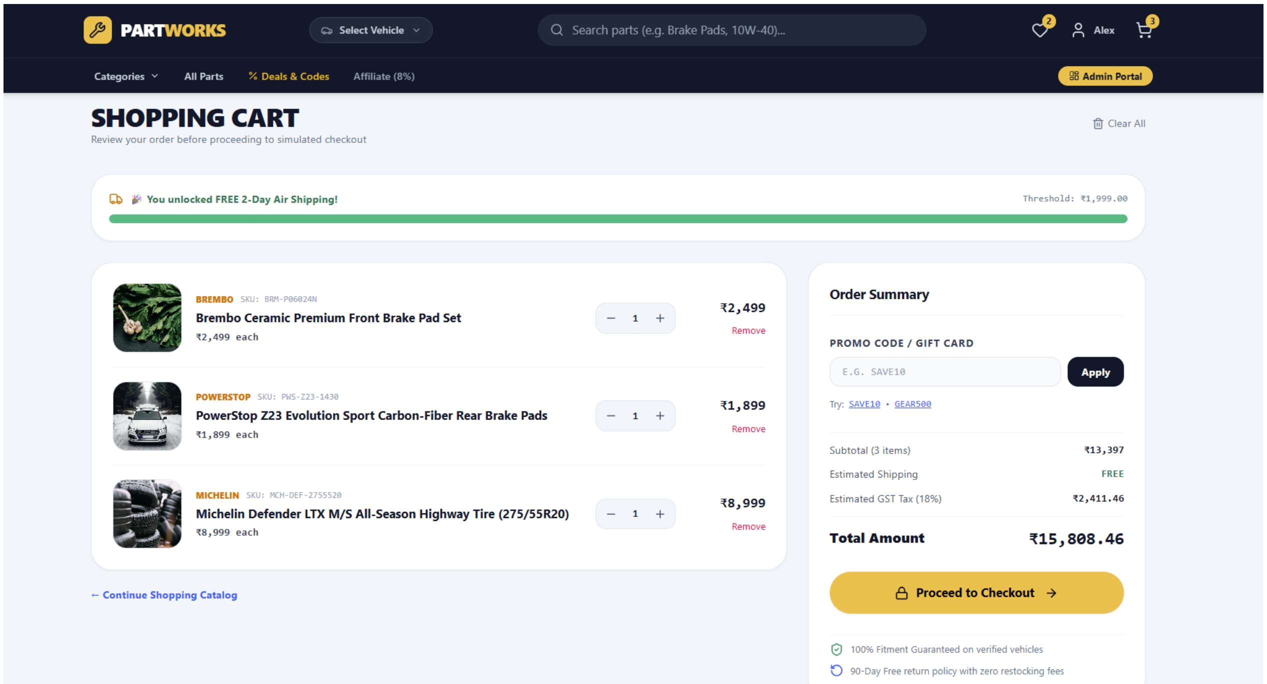
HOME PAGE



ADMIN PORTAL



SHOPPING CART



6.1 Test Plan/ Test Cases

TC01 – Load homepage: verify the homepage loads with navigation and featured products. Expected: Homepage loads correctly with navigation and featured products visible.

TC02 – Browse catalog: verify the catalog page displays parts with working filters. Expected: Catalog page displays list of parts with filters applied correctly.

TC03 – Use vehicle selector: verify correct compatible parts are shown for a selected vehicle. Expected: Correct compatible parts shown for selected vehicle.

TC04 – Add item to cart: verify item appears in cart with correct quantity/price. Expected: Item appears in cart correctly.

TC05 – Checkout flow: verify user can complete checkout and reach order confirmation. Expected: Checkout completes and order confirmation is shown.

TC06 – User login: verify valid credentials log the user in and invalid credentials show an error. Expected: Login behaves correctly for both cases.

TC07 – Admin product management: verify admin can add, edit, and remove a product. Expected: Product changes are reflected correctly.

TC08 – Responsive design check: verify layout adapts correctly on mobile and desktop screen sizes. Expected: Layout is responsive and usable on all tested screen sizes.

6.2 Test Procedure

Each test case was executed manually by performing the corresponding user action in the running application (in a local development environment) and comparing the actual behavior against the expected result. Core flows — browsing, vehicle selection, cart, checkout, login, and admin actions — were tested on both desktop and mobile browser widths to confirm responsive behavior.

6.3 Performance Outcome

All eight test cases (TC01–TC08) passed successfully. The homepage, catalog, and vehicle selector loaded and functioned correctly, the cart and checkout flow worked end-to-end through to order confirmation, login handled both valid and invalid credentials correctly, the admin panel supported product management as expected, and the layout adapted correctly across mobile and desktop screen sizes.

7 My learnings

This internship gave me hands-on experience that went well beyond classroom learning. Building PartWorks within a 6-week timeline taught me how to plan a project properly before writing code — breaking it down into pages and components rather than building everything in one place. I became much more comfortable with React and TypeScript, particularly in structuring a larger project with multiple pages and shared state managed through Context API. I also learned to think through the full user journey of an e-commerce platform, from browsing and searching for a product to completing checkout, rather than building isolated features in isolation. Working on the admin panel gave me a chance to design from the business owner's perspective as well as the customer's, which was a different and valuable way of thinking about the product. More generally, this internship gave me a sense of how real industrial projects work — working to a fixed deadline, tackling a problem statement fairly independently, and documenting the work properly in a report — all of which I expect to carry directly into my career.

8 Future work scope

While PartWorks currently covers the core features of an e-commerce platform, there is scope to extend it further: integrating a real backend (Node.js/Express) and database (such as MongoDB or PostgreSQL) to replace the current mock data files; adding a real payment gateway (such as Razorpay or Stripe) to support live transactions instead of a simulated checkout; allowing customer reviews and ratings on parts; adding real-time order status tracking; improving the vehicle selector with smarter filtering and personalized recommendations based on browsing history; building a mobile app version for wider reach; and adding an analytics dashboard to the admin panel to support better business decisions. With these additions, PartWorks could evolve from a functional prototype into a production-ready e-commerce platform for the automotive parts industry.